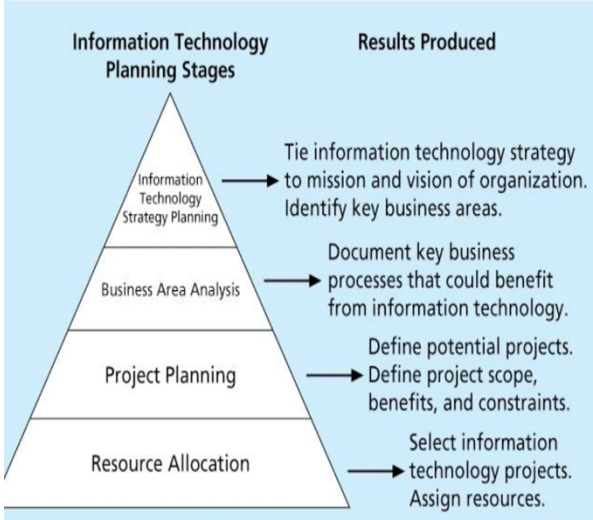
Lecture 3

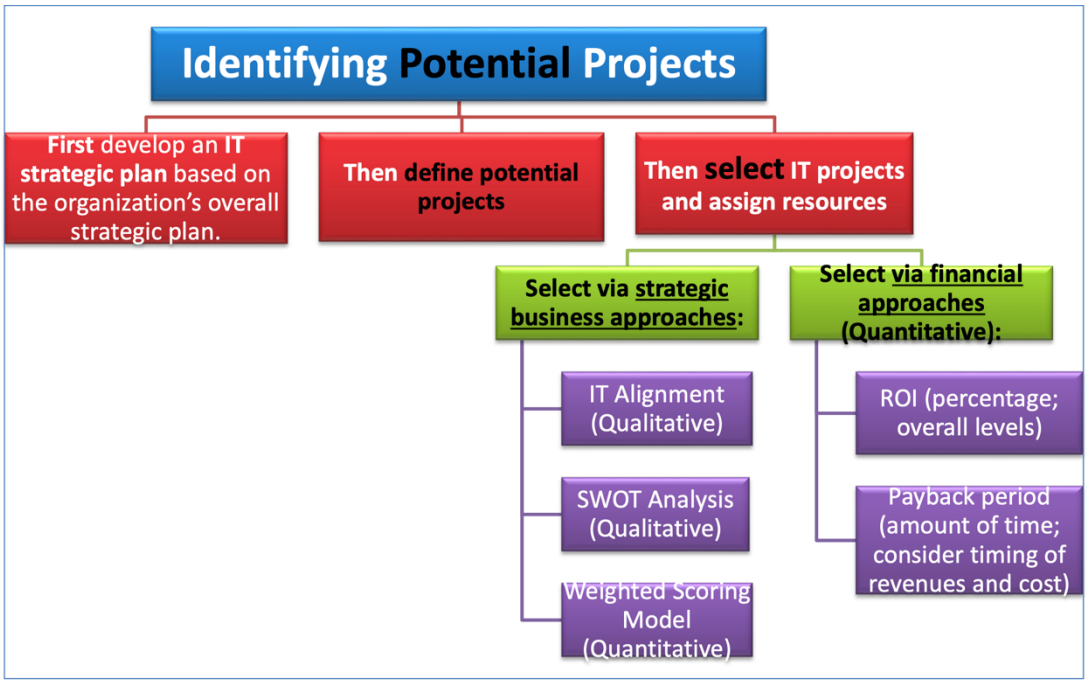
Part 1: Selecting IT Projects

- **Project Portfolio Management**
  - **Definition**
    - **helps organizations** manage projects and programs as a **strategic investment portfolio**.
    - **Have strategical goal (long term for overall goals) unlike Project management that have tactical goals (short term specific)**
      - **Example: become data-driven company or e-commerce business**
    - **Goal is to** Optimize resource allocation and ensure **high-impact projects** get executed

  - **Portfolio Manager**
    - ensure **projects align with business goals**
    - They **analyze, prioritize, and select** projects based on **strategic value and ROI**
- **IT Planning Process**



- **Program**
  - **Definition**
    - A group of related projects managed in a coordinated way to obtain benefits and control for strategic goals .
  - **Program manager**
    - oversee programs; often act as **boss** for **project managers**
    - **Should have technical knowledge , leadership skills, stakeholder management skills**



- **Strategic Planning**
  - involves determining long-term objectives, predicting future trends, and projecting the need for new products and services

- **Define Potential Projects**

- **Select IT Projects**
  - There are usually more projects than available time and resources to implement them
  - Methods for selecting

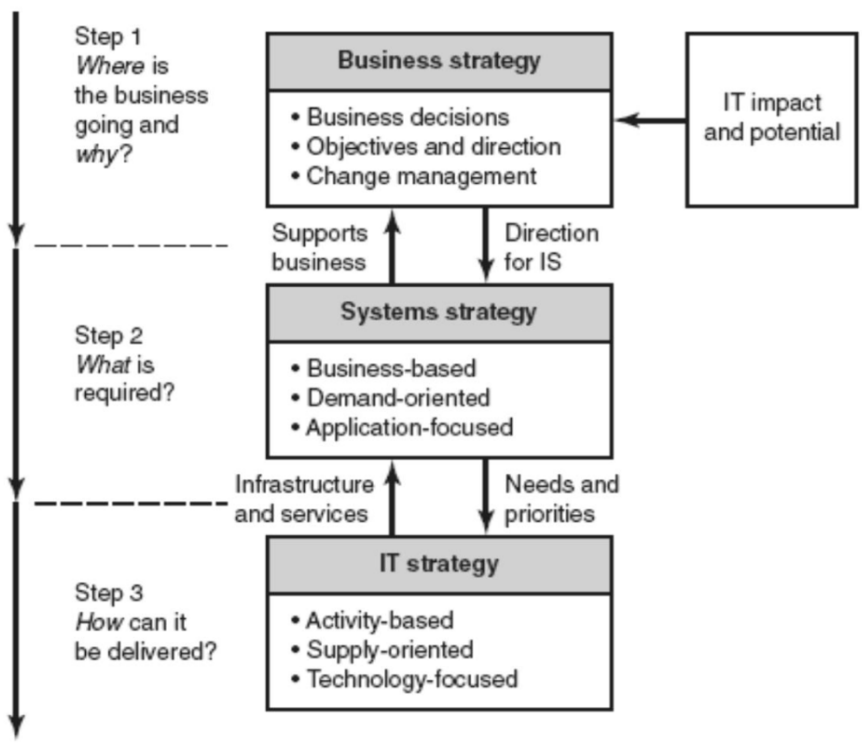
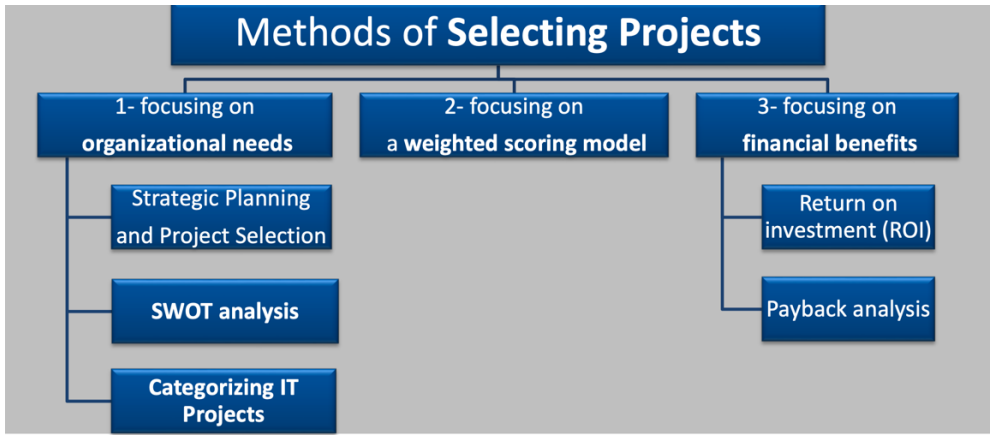


FIGURE 4-2 Traditional Strategy Making

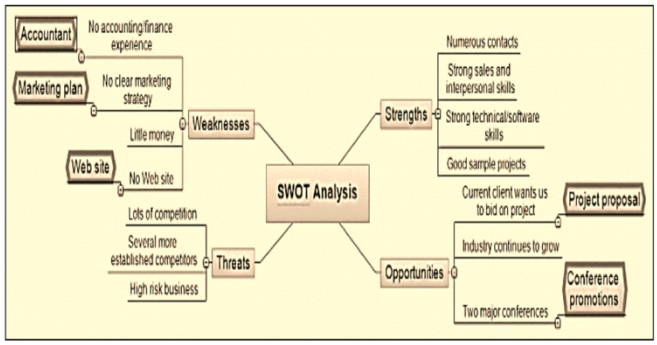


1. **Focusing on organizational needs**

   [1.1] Strategic Planning and Project Selection

   - **planning** involves determining long-term objectives, predicting future trends, and projecting the need for new products and services
   - Organizations identify **potential projects,** use realistic **methods to select which projects** to work on , formalize project **initiation by issuing a project charter**

   [1.2] SWOT Analysis (Strength, Weakness, Opportunity, Threat)



   [1.3] Categorizing IT Projects

   - One categorization is whether it address a problem an opportunity or a directive
   - Another one how long does it take and when is it needed
   - Another is overall priority of project

2. **Focusing on weighted scoring model (Systematic process)**

   1) Identify criteria
   2) Assign weight to each criterion
   3) Assign score for each criterion for each project
   4) Multiply scores by weights

   - The higher the score the better (range 0-100)

3. **Focusing on financial benefits**

[3.1] Return of Investment (ROI)

- **ROI =[(total benefits - total costs) / total costs] x 100**
- **The higher ROI the better**

[3.2] Payback Analysis (Breakeven)

- **Payback occurs when total benefits** received **equal** the **initial cost.**
- **Methods (Exercise in Slide 23!!!)**
  1) **Averaging method**
     - **Formula: Initial Investment ÷ Annual Cash Inflows**
     - **Example:** A business spends **$12,000** on software and earns **$3,000 yearly** →Payback = **4 years** ($12,000 ÷ $3,000).
  2) Subtraction Method
     - **Process:** Subtract each year's cash inflow from the initial cost until full recovery.
     - **Example:**
       - Year 1: Cash inflow = **$2,000** → Remaining = **$8,000**
       - Year 2: Cash inflow = **$3,000** → Remaining = **$5,000**
       - Year 3: Cash inflow = **$5,000** → Remaining = **$0**
       - (Payback achieved in 3 years)

Part 2: IT Project Integration Management

1. Initiating

- Process: Develop project charter (a **document** that **formally recognizes the existence of a project** and provides direction on the project's objectives and management.)
- Output: Project charter

2. Planning

- Process: Develop project management plan
- Output: team contract, scope statement, work breakdown structure (WBS), project schedule with all dependencies and resources, list of prioritized risks

3. Executing

- Process: Direct and manage project work
- Outputs: Deliverables, work performance data, change requests, project management plan updates, project documents
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4. Project Monitoring and Controlling

- Process: Monitor and control project work
- Outputs: Change requests, project management plan updates, project documents updates
- Process: Perform integrated change control
- Outputs: Approved change requests, change log, project management plan updates, project documents updates

5. **Project Closing**

Process: Close project or phase (involves gaining **stakeholder and customer acceptance**)

- Outputs: Final product, service, or result transition; organizational process assets updates

Part 3: IT Project Procurement Management

Procurement means acquiring goods and/or services(by client) from an outside source (Vendor) -- Other terms include purchasing and outsourcing

1. **Procurement Planning** **(Example on Slide 46!!!!)**
   - Determining (whether/ what/ when/ how/ how much) to procure
   - 4 types of contracts
     1) **Fixed Price or lump-sum contracts**
        - **Firm Fixed Price (FFP) ->** Price is set and does not change
        - **Fixed Price with Economic Price Adjustment (FPEPA) ->** Price adjusts based on economic conditions.
        - **Fixed Price Incentive Fee (FPIF) ->** Fixed price with a performance-based bonus.
          - It **motivates the supplier** to perform better (finish faster reduce delays).

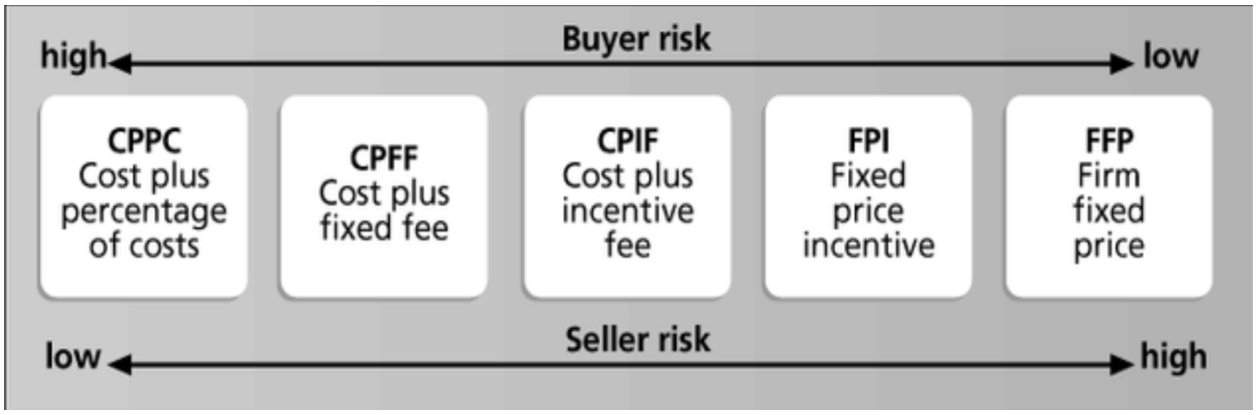
     2) **Time and material contracts (hybrid between Fixed [1] & Cost-Re[3])**
        - **Time-Based ->** Buyer pays for the seller's time spent.
        - **Material-Based ->** Buyer reimburses actual material cost
          - **Example:** A repair service charges for labor **plus** the cost of replacement parts.

     3) **Cost-Reimbursable contracts** **(Example on Slide 51!!!!)**

        (Note!!) Reimbursed costs means actual cost of expenses
        - **Cost Plus Incentive fee (CPIF) ->** Reimbursed costs **plus** an incentive for cost savings/performance.
          - **Has fixed fee + Incentive bonus** **(see example in slide 51)**
          - **Example:** A research project receives funding for actual costs plus an incentive if it stays under budget.
        - **Cost Plus Fixed Fee ->** Reimbursed costs **plus** fixed fee
        - **Cost Plus Percentage of Costs ->** Reimbursed costs **plus** a percentage of actual costs.

     4) **Unit price contracts**
        - Fixed Price Per Unit -> Payment is based on a fixed price per unit of service.
          - **Example:** A contractor charges $100 per meter for installation.

- **Contract Types vs Risk**



- **Statement of Work (SOW) -> description** of the work **required for the procurement**

I.	**Scope of Work:** Describe the work to be done to detail. Specify the hardware and software involved and the exact nature of the work.
II.	**Location of Work:** Describe where the work must be performed. Specify the location of hardware and software and where the people must perform the work
III.	**Period of Performance:** Specify when the work is expected to start and end, working hours, number of hours that can be billed per week, where the work must be performed, and related schedule information.
IV.	**Deliverables Schedule:** List specific deliverables, describe them in detail, and specify when they are due.
V.	**Applicable Standards:** Specify any company or industry-specific standards that are relevant to performing the work.
VI.	**Acceptance Criteria:** Describe how the buyer organization will determine if the work is acceptable.
VII.	**Special Requirements:** Specify any special requirements such as hardware or software certifications, minimum degree or experience level of personnel, travel requirements, and so on.

2. **Solicitation Planning (Preparing several documents)**
   - **Request for Proposals (RFPs):** used to **solicit proposals** from **prospective** sellers
     - **Template:**

I.	Purpose of RFP
II.	Organization's Background
III.	Basic Requirements
IV.	Hardware and Software Environment
V.	Description of RFP Process
VI.	Statement of Work and Schedule Information
VII.	Possible Appendices
	A. Current System Overview
	B. System Requirements
	C. Volume and Size Data
	D. Required Contents of Vendor's Response to RFP
	E. Sample Contract

   - **Requests for Quotes (RFQs):** used to **solicit quotes for well-defined procurements**
   - **Invitations for bid or negotiation** and **initial contractor responses** are also **part of solicitation planning**

3. **Solicitation**
   - Solicitation involves **obtaining proposals** or **bids** from **prospective sellers**

4. **Source Selection**
   - **evaluating bidders' proposals -> choosing the best one-> negotiating** the contract **-> awarding** the contract

Sample Proposal Evaluation Sheet

		Proposal 1		Proposal 2		Proposal 3	
Criteria	Weight	Rating	Score	Rating	Score	Rating	Score
Technical Approach	30%						
Management Approach	30%						
Past Performance	20%						
Price	20%						
Total Score	100%						

Detailed Criteria for Selecting Suppliers

Criteria	Possible Points	Supplier 1 Points	Supplier 2 Points	Supplier 3 Points
Project manager s educational background and experience	10	8	6	9
Project manager is PMP certified	5	5	0	5
Presentation on management approach	5	4	3	5
Organization s project management methodology	10	7	4	9
Total Score	30	24	13	28

5. **Contract Administration**
    - ensures that the seller's performance **meets contractual requirements**
    - **Changes** must be **documented** in writing.
    - Contract close-out includes
        i. **product verification** to determine if all work was completed correctly and satisfactorily
        ii. **administrative activities** to update records to reflect results
        iii. **archiving information** for future use